

Data Analyst Case Study : Data Analysis and Dashboard creation

Task: Use the shared dummy data to perform data analysis and create dashboards for better understanding

Note : Attach the final code for the dashboards created, python / notebook files with the analysis. No need for PPT. Make sure your code is working properly, and specify the python version, package versions being used for the exercise.

Data:

The shared data represents the Profit and Loss for a cloud kitchen company across different kitchen stores, cities and the corresponding values of the fixed and operating costs, along with the expenses happening at the monthly level.

We want you to explore the data for analysis purposes, and make use of open-source tools, like Plotly or Streamlit, and create dashboards for it by applying different types of filters on the below shared list of columns.

The exercise involves 2 parts :

Dashboard 1 (Kitchen Level PNL) : Create kitchen level Profit & Loss information by putting filters on the shared dataset. The list of the columns to be used for filtering the overall data :

- STORE,GM% GM,CM %,CM , EBITDA ,NET_REVENUE, Month, REVENUE COHORT, CM COHORT, EBITDA CATEGORY,EBITDA COHORT

Dashboard 2 (VARIANCE Level PNL) : Make use of the 'VARIANCE' column to create a second dashboard by creating buckets on the basis of the value segments shared in the image below. This dashboard is to be split into 2 parts :

- a. Sub-dashboard 1 : Create a summary of the average variance % of the kitchens under specific revenue (NET_REVENUE column) categories
- b. Sub-dashboard 2 : Create a summary of the total number of kitchen stores on the basis of the Variance filter used at the top, and then further bifurcate the revenue (NET_REVENUE column) field in the shared value ranges. Populate the table with the count of kitchen stores (STORE column) for each of the respective months

Notes :

*The revenue category filter is cohort-based (fixed) .Additionally, it should include range-based filters for EBITDA, Contribution Margin (CM), and Revenue.

*Variance : Wastage of food material

Proposed steps:

1. Create dataset in the desired format for both the dashboards
2. Develop Streamlit / Plotly app with the

3. Include code segments to optimize the performance of the dashboard, in case the data is getting refreshed on real time basis
4. Bonus points for creating fancy features in the dashboard, like the last image shared in the document below

1. Kitchen Level PNL -

Select EBITDA Range (in ₹)



Zone

EBITDA category

KITCHEN SNAPSHOT

Store

Month

Revenue category

Mar 1, 2025												Feb 1, 2025		J...	
Store	Net reven...	GM %	CM %	EBITDA	EBITDA %	Net reven...	GM %	CM %	EBITDA	EBITDA %	Net r				

2. VARIANCE Level PNL –

- a. 1st sub-dashboard Visual -Store %
- b. 2nd sub-dashboard Visual - Store Count

VARIANCE BY REVENUE CATEGORY

The tables below summarise the average variance % on cart of the kitchens under revenue categories and the count of kitchen under each revenue category to further drill down on the variance category

Revenue category	Mar 2025	Feb 2025	Jan 2025	Dec 2024	Nov 2024	Oct 2024	Sep 2024
(a) Below INR 15 lacs							
(b) INR 15 to 25 lacs							
(c) INR 25 to 35 lacs							
(d) INR 35 to 45 lacs							
(e) Above INR 45 lacs							
Grand total	1.3%	1.5%	1.9%	1.3%	1.6%	1.3%	1.5%

Revenue category	Mar 2025	Feb 2025	Jan 2025	Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024
(a) Below INR 15 lacs										
(b) INR 15 to 25 lacs										
(c) INR 25 to 35 lacs										
(d) INR 35 to 45 lacs										
(e) Above INR 45 lacs										
Grand total	353	351	350	346	340	336	338	337	336	336

☒ Variance category

🔍 Type to search

☒ (a) Var < 2%

☒ (b) Var 2% to 3%

☒ (c) Var 3% to 5%

☐ Variance category (1)

🔍 Type to search

☐ (a) Var < 2%

☒ (b) Var 2% to 3%

ONLY

☐ (c) Var 3% to 5%

only

☐ (d) Var > 5%

The final analysis of the data should be represented in a Streamlit or Plotly dashboard in the shared template. In addition to this, if there are any other level of insights you are able to gather from the shared data then please share those as well. Preferred language is Python.